

### **Lab #3: Preparing a Requirements Document for AI Coding Agents**

#### ***Overview:***

LessonLoop is a web application for early-career teachers who need a faster, more organized way to build lessons, differentiate activities, and keep classroom planning materials in one place. The goal of this document is to define the product clearly enough that an AI coding agent can use it as a reliable blueprint for planning, building, and testing the first usable version of the product.

The problem solves a bunch of common workflow issues. Many new teachers plan lessons across scattered documents, notes apps, curriculum pages, and browser tabs. This makes it hard to reuse ideas, adapt for diverse learners, and prepare quickly for practicum or classroom teaching. LessonLoop brings these tasks into one workspace where users can create lesson plans from templates, align them to curriculum outcomes, adapt activities for different learner needs, and export clean teaching documents.

This first release should focus on a practical, stable core experience rather than trying to do everything. The product should feel simple, fast, and trustworthy. It should help a teacher move from a rough lesson idea to a classroom-ready plan without needing advanced technical knowledge.

#### ***Product Vision***

Build a planning tool that uses AI to help teachers turn ideas into a clear, differentiated, and exportable lesson plan as efficiently as possible.

#### ***Problem Statement***

1. Lesson-planning content is spread across multiple tools
2. Differentiation is often added too late to be used from the beginning
3. The curriculum is difficult to track consistently, especially if a teacher is teaching more than one class
4. Finished plans are hard to format cleanly for real classroom use

Early career teachers often face the listed problems above. As a result, teachers spend too much time figuring out how to set up many meticulous parts of a whole piece, causing their teaching to be at risk.

#### ***Target Users***

Primary users will consist of early teachers between the ages of 20 and 30, but this tool will not be limited to just new teachers; it will also adapt to more experienced teachers as well. The goal of this tool is to also increase efficiency for teachers, making it workable software for any user.

### ***Core Features***

1. **Structured lesson builder:** Users can create a lesson using required sections: title, grade level, subject, duration, learning objectives, materials, lesson sequence, assessment, differentiation, and reflection notes.
2. **Curriculum alignment:** users can attach curriculum outcomes or standards to each lesson. In version 1, this can be supported with a searchable manual entry system or a small library instead of full integration.
3. **Differentiation Planner:** The lesson editor should include dedicated prompts for supports, extensions, accommodations, and multiple ways students can engage or show learning.
4. **Reusable templates:** Users can start from blank, sample, or previously saved templates. Templates reduce planning time and help with consistency.
5. **Export-Ready Formatting:** Users can export a clean, readable lesson plan as a PDF or DOCX for practicum submission, sharing, or printing.
6. **Lesson Archive:** Users can view, search, duplicate, archive, or delete prior lesson plans from a dashboard

### ***User Journey***

A new user signs up, lands on a dashboard, selects 'Create lesson,' chooses either a blank plan or a template, fills out the guided sections, tags curriculum outcomes, adds differentiation and assessment notes, previews the lesson, exports it, and later returns to duplicate or revise it for another class

### ***Technical Requirements***

Platform:

- web application accessible on laptop and tablet

Data storage:

- stores user-created lesson plans and related data

Interface requirements:

- Clean dashboard, calendar view, task input form

Functional requirements:

- Authentication and account management
- Dashboard and lesson library
- Lesson editor
- Curriculum alignment
- Differentiation assessment
- Export and document output

Non-Functional Requirements:

- Good connection
- Autosave

- Adapt to screen sizes (laptop and tablet)
- Accessibility for colour, keyboards, and labels
- Encrypted data
- Properly exported data

## ***User Stories***

### **Story 1: Create an accounts**

- User story: as a new user, I want to create an account so that my lesson plans are privately and can be accessed later

#### Acceptance Criteria:

- User can sign up with name, email, and password
- The user sees validation errors for invalid email or weak password
- A successful sign-up redirects user to the dashboard

### **Story 2: Log in Securely**

- User story: as a returning user, I want to log in securely so that I can continue working on my saved lesson plans.

#### Acceptance Criteria:

- User can log in with valid credentials
- The user sees an error message for invalid credentials
- User sessions persist after refresh until logout or session expiry

### **Story 3: Create a Lesson From Scratch**

- User story: as a preservice teacher, I want to create a lesson from a guided template so that I do not forget essential planning elements

#### Acceptance Criteria:

- User can open a new lesson from the dashboard
- The editor displays all required planning sections
- Required fields are visually marked
- Lesson draft autosaves while the user types

### **Story 4: Create a Lesson From a Template**

- User story: As a busy teacher, I want to start from a reusable template so that I can plan more quickly

#### Acceptance Criteria:

- User can select a template from the dashboard
- A new editable lesson is created without changing the original template
- Template-derived lessons can be saved and exported normally

**Google Stitch Link:** <https://stitch.withgoogle.com/projects/8973034767123117819>

Prompt:

Create three clean web app mockups for a lesson planning tool called LessonLoop designed for early-career teachers.

Screen 1: a dashboard showing saved lessons, templates, and a 'Create Lesson' button.

Screen 2: lesson editor with sections for objectives, materials, lesson sequence, assessment, curriculum alignment, and differentiation supports.

Screen 3: a lesson preview and export screen with a formatted layout and options to download as PDF or DOCX.

Use a clean, professional education style with blue, white, and soft grey colours.